

- 1 (a)(i) For a fixed mass of ideal gas, pV is proportional to T .

$$\text{Using } p_A V_A = nRT$$

C1

$$T = \frac{400 \times 0.30}{0.050 \times 8.31} = 289 \text{ K}$$

A1

(a)(ii) Kinetic energy = $\frac{3}{2} nRT = (\frac{3}{2})(0.050)(8.31)(289)$
 $= 180 \text{ J}$

A1

(a)(iii) Work done by the gas = area under the graph

$$= (8.5 \text{ big squares})(100 \times 10^3)(0.20 \times 10^{-3})$$

M1

$$= 170 \text{ J (allow 160-180J)}$$

A1

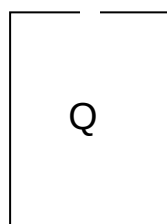
(a)(iv) Since temperature of the ideal gas is constant, increase in internal energy $\Delta U = 0$. As the gas expands from A to B, work done on the gas W is negative.

M1

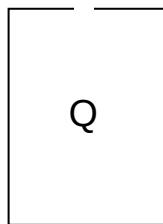
Using first law of thermodynamics, $\Delta U = Q + W$, the heat supplied to the gas Q is positive. Hence, heat flows into the gas.

A1

(b)



$p_0, V, n, 300 \text{ K}$



$p_0, V, n', 430 \text{ K}$

$$\text{Initial temperature} = 27 + 273 = 300 \text{ K}$$

$$\text{Final temperature} = 157 + 273 = 430 \text{ K}$$

C1

$$p_0 V = nR \times 300$$

$$p_0 V = n' R \times 430$$

M1

$$\frac{n'}{n} = \frac{300}{430} = 0.698$$

A1

2

2(a)

The box is initially at rest. The tension in the spring initially balances the total